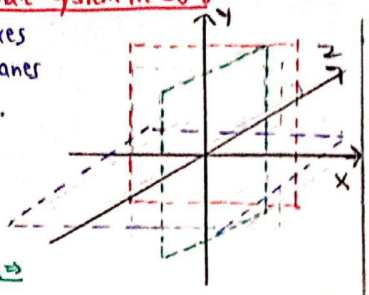
3D-Geometry-1

Cartesian Coordinate System in 3D :

- 3 coordinate axes
- 3 coordinate planes
- 8 ~~quadrants~~ octants.

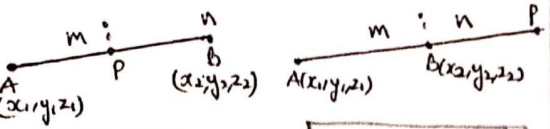


Distance Formula ⇒

Distance b/w $A(x_1, y_1, z_1)$ & $B(x_2, y_2, z_2)$ is

$$AB = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$$

Section Formula ⇒

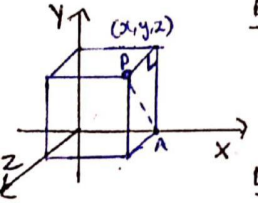


Internal	External
$x = \frac{mx_2+nx_1}{m+n}$	$x = \frac{mx_2-nx_1}{m-n}$
$y = \frac{my_2+ny_1}{m+n}$	$y = \frac{my_2-ny_1}{m-n}$
$z = \frac{mz_2+nz_1}{m+n}$	$z = \frac{mz_2-nz_1}{m-n}$

Coordinates of Centroid of a △ ⇒

$$G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}, \frac{z_1+z_2+z_3}{3}\right)$$

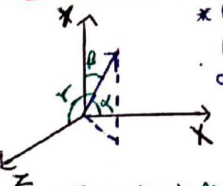
Distance of a point from coordinates planes & axes ⇒



Distance of P from
- xy plane = |z|
- yz plane = |x|
- zx plane = |y|

Distance of P from (obtained by applying Pythagoras theorem on right angled △s.)
- x-axis (PA) = $\sqrt{y^2+z^2}$
- y-axis (PB) = $\sqrt{z^2+x^2}$
- z-axis (PC) = $\sqrt{x^2+y^2}$

Direction Cosines (DCs) :



* DCs are cos of angles made by the vector with coordinate axes.

$\cos\alpha, \cos\beta, \cos\gamma \Rightarrow$ DCs.
(l) (m) (n)

* Important Result ⇒

1) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = l^2+m^2+n^2 = 1$
2) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Vector formed by DCs is always a unit vector $(l\hat{i} + m\hat{j} + n\hat{k})$

Direction Ratios and Direction Vectors :

→ DRs are multiples of DCs and are infinite in number.

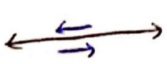
$$DRs = \lambda(DCs)$$

$$(a,b,c) = \lambda(l,m,n) \quad (\lambda \to \text{constant})$$

→ Vector formed by dir^n ratios is called 'Direction Vector'

Eg: $\vec{v} = a\hat{i} + b\hat{j} + c\hat{k}$
$= \lambda l\hat{i} + \lambda m\hat{j} + \lambda n\hat{k}$

→ For a line, there are two DCs.



Eg: $L: 2\hat{i} - 2\hat{j} + \hat{k}$

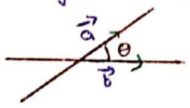
Sol^n Unit vector $= \pm\left(\frac{2\hat{i}-2\hat{j}+\hat{k}}{3}\right)$

$\Rightarrow$ DCs $= \pm\left(\frac{2}{3}, \frac{-2}{3}, \frac{1}{3}\right)$ Ans

NOTE: To get DCs of any line/vector, we need to divide that vector by its magnitude.

Angle b/w two lines :

It is the acute angles b/w vectors along the intersecting lines.



$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$$

Q Find angle b/w lines whose DCs satisfy the equation $l+m+n=0$ and $l^2 = m^2+n^2$.

Sol^n We know, $l^2+m^2+n^2 = 1$
$l+m+n = 0$
$l^2 = m^2+n^2$
} 3 equations, 3 variables

$l^2 + l^2 = 1 \Rightarrow 2l^2 = 1 \Rightarrow \boxed{l = \pm\frac{1}{\sqrt{2}}}$

Now, $l = -m-n$, put this in eq^n ①.

$(-m-n)^2 = m^2+n^2$

$m^2+n^2+2mn = m^2+n^2$

$\Rightarrow \boxed{2mn = 0}$

When $m=0$: $n=-l$, DCs $= \left(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}\right)$

When $n=0$: $m=-l$, DCs $= \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right)$

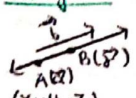
$$\cos\theta = \frac{\left(\frac{1}{2}+0+0\right)}{(1)(1)} \Rightarrow \frac{1}{2}$$

$\Rightarrow \boxed{\theta = \pi/3}$ Ans

Equation of Straight Lines :

To write equation of a line, we need a vector along the line and a point on it.

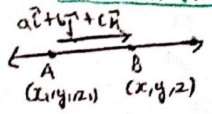
Vector form



$\vec{b} - \vec{a} = \lambda\vec{b}$

$\Rightarrow \boxed{\vec{r} = \vec{a} + \lambda\vec{b}}$

Cartesian Form



$\because \vec{AB} \parallel (a\hat{i}+b\hat{j}+c\hat{k})$

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = \lambda$$

Parametric Coordinates of a Line :

Parametric coordinates are simply values of x, y & z in terms of λ.

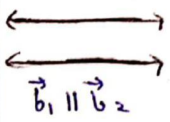
Eg: $\frac{x+3}{3} = \frac{y-1}{6} = \frac{z+3}{4} = \lambda$

Sol^n $x = 3\lambda-3$, $y = 5\lambda+1$, $z = 4\lambda-3$

Eg: $\vec{r} = \hat{i} + \hat{j} + \lambda(2\hat{i} - \hat{j} + \hat{k})$

Sol^n $x = 1+2\lambda$, $y = 1-\lambda$, $z = 0+\lambda$

Condition of Parallel and Iar Lines :



$\Rightarrow \boxed{\vec{b_1} = \lambda\vec{b_2}}$



$\vec{b_1} \perp \vec{b_2}$

$\Rightarrow \boxed{\vec{b_1}\cdot\vec{b_2} = 0}$

Point of intersection of two Lines :

→ Write parametric coordinates for both lines and equate corresponding points.

→ Put the value of unknown parameters in one of the equations formed by equating parametric points.

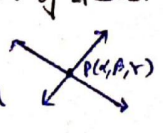
If eq^n is satisfied (P.O.I exist)

If eq^n is not satisfied (P.O.I does not exist)

Q Find the point of intersection of L_1 & L_2.

L_1: $\frac{x-2}{6} = \frac{y-1}{-8} = \frac{z+2}{10} = \lambda$

L_2: $\frac{x+1}{6} = \frac{y+10}{7} = \frac{z-5}{-2} = \mu$



Sol^n $P_1 \equiv (6\lambda+2, -8\lambda+1, 10\lambda-2)$
$P_2 \equiv (6\mu-1, 7\mu-10, -2\mu+5)$

$6\lambda+2 = 6\mu-1$

$6(\lambda-\mu) = -3 \Rightarrow \boxed{\lambda-\mu = -\frac{1}{2}}$ — ①

$10\lambda - 2 = -2\mu + 5 \Rightarrow 10\lambda + 2\mu = 7$

$\Rightarrow \boxed{5\lambda + \mu = \frac{7}{2}}$ — (2)

(1) + (2) $\Rightarrow 6\lambda = \frac{7}{2} - \frac{1}{2} \Rightarrow 6\lambda = 3$

$\Rightarrow \boxed{\lambda = \frac{1}{2}}$ $\boxed{\mu = 1}$

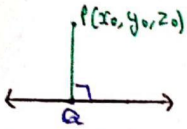
3rd equation →

$-8\lambda + 1 = 7\mu - 10$

$-8\left(\frac{1}{2}\right) + 1 = 7(1) - 10 \Rightarrow (-3 = -3)$ ✓

$P \equiv (6\lambda - 1,\ 7 - 10,\ -2 + 5) \equiv (5, -3, 3)$ Ans

Foot of Perpendicular from Point :



Method ⇒

① Assum foot of ⊥ar (Q) on the line as parametric coordinates.

$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

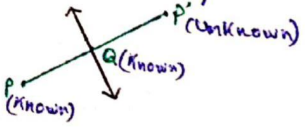
② Now put,

$\boxed{(\vec{PQ}) \cdot (\text{Dir}^n \text{ vector of line}) = 0}$

This will give us value of assumed parameter and hence the point Q will be obtained

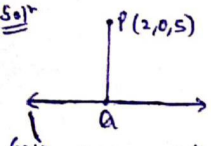
Image of a Point →

For Image of point P, simply find its foot of ⊥ar (Q) and apply mid-point formula b/w P & P' with Q as mid point.



Ques The foot of ⊥ar of the point (2,0,5) on the line $\frac{x+1}{2} = \frac{y-1}{5} = \frac{z+1}{-1}$ is (α, β, γ). Then find $\frac{\beta}{\gamma}$.

Sol^n



$Q \equiv (2\lambda - 1,\ 5\lambda + 1,\ -\lambda - 1)$

$\vec{PQ} = (2\lambda - 3,\ 5\lambda + 1,\ -\lambda - 6)$

$\left(\frac{x+1}{2} = \frac{y-1}{5} = \frac{z+1}{-1}\right) = \lambda$ Dir^n vector of line $(\vec{N}) = (2, 5, -1)$

Now, $\vec{PQ} \cdot \vec{N} = 0$

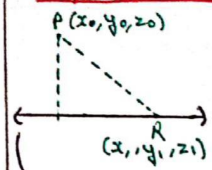
$(4\lambda - 6) + (25\lambda + 5) + (\lambda + 6) = 0$

$30\lambda + 5 = 0 \Rightarrow \boxed{\lambda = -1/6}$

So, $\frac{\beta}{\gamma} = \frac{5\left(\frac{-1}{6}\right) + 1}{-\left(\frac{-1}{6}\right) - 1} \rightarrow \frac{1/6}{-5/6}$

$\boxed{\beta/\gamma = \frac{-1}{5}}$ Ans

Perpendicular Distance of Point from Line :



$\text{⊥ar distance} = \frac{|\vec{PR} \times \vec{b}|}{|\vec{b}|}$

$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

$\vec{b}$ = Dir^n vector of line

R = Point on line

Ques The distance of the point P(4,6,−2) from the line passing through the point (−3,2,3) and parallel to a line with direction ratios 3,3,−1 is equal to ____.

Sol^n $P \equiv (4,6,-2)$ $\quad \vec{PR} = (-7, -4, 5)$

$R \equiv (-3,2,3)$

$\vec{b} = (3,3,-1)$

$$\vec{PR} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -7 & -4 & 5 \\ 3 & 3 & -1 \end{vmatrix}$$

$= \hat{i}(4-15) - \hat{j}(7-15) + \hat{k}(-21+12)$

$= -11\hat{i} + 8\hat{j} - 9\hat{k}$

So, (PQ) ⊥ar distance $= \frac{|\vec{PR} \times \vec{b}|}{|\vec{b}|} = \frac{\sqrt{11^2 + 8^2 + 9^2}}{\sqrt{3^2 + 3^2 + 1^2}}$

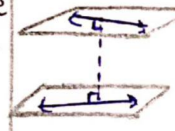
$PQ = \sqrt{\frac{121 + 64 + 81}{19}} = \sqrt{\frac{266}{19}}$

$\boxed{PQ = \sqrt{14}}$ Ans

Skew Lines :

↳ Non parallel, non intersecting lines are called skew Lines.

Shortest Distance b/w two skew lines :→



$\text{S.D.} = \left|\frac{(\vec{a_2} - \vec{a_1}) \cdot (\vec{b_1} \times \vec{b_2})}{|\vec{b_1} \times \vec{b_2}|}\right|$

[This is modulus, not magnitude]

* If S.D. = 0 then it implies that lines are intersecting.

Shortest Distance b/w two Parallel Lines :

$\vec{r} = \vec{a_1} + \lambda\vec{b}$

$\vec{r} = \vec{a_2} + \mu\vec{b}$

$\text{S.D.} = \frac{|(\vec{a_2} - \vec{a_1}) \times \vec{b}|}{|\vec{b}|}$

Ques The shortest distance between the lines $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ is ____.

Sol^n $a_1 = (-2, 0, 5)$

$a_2 = (4, 1, -3)$

$\Rightarrow (a_2 - a_1) = (6, 1, -8)$

$$\vec{b_1} \times \vec{b_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 1 & 2 & 0 \end{vmatrix} = \hat{i}(0-4) - \hat{j}(0-2) + \hat{k}(2+2)$$

$\vec{b_1} \times \vec{b_2} = -4\hat{i} + 2\hat{j} + 4\hat{k}$ (Only here)

$= (-4, 2, 4)$

$= (-2, 1, 2)$

In calculation of $\vec{b_1} \times \vec{b_2}$, we can divide resultant by some constant if we want.

$\text{S.D.} = \left|\frac{(\vec{a_2} - \vec{a_1}) \cdot (\vec{b_1} \times \vec{b_2})}{|\vec{b_1} \times \vec{b_2}|}\right|$

$= \left|\frac{(6,1,-8) \cdot (-2,1,2)}{\sqrt{9}}\right| = \left|\frac{-12 + 1 - 16}{3}\right|$

$= |-9| = 9$ Ans

Ques Let S be the set of all values of λ, for which the S.D. b/w lines $\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ and $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ is 13. Then $8\sum|\lambda|$ is equal to ____.

Sol^n $a_1 = (\lambda, 3, -6)$

$a_2 = (-\lambda, 0, 6)$

$\Rightarrow (a_2 - a_1) = (-2\lambda, -3, 12)$

$$\vec{b_1} \times \vec{b_2} = \begin{vmatrix} i & j & k \\ 0 & 4 & 1 \\ 3 & -4 & 0 \end{vmatrix} = \hat{i}(0+4) - \hat{j}(0-3) + \hat{k}(0-12)$$

$\vec{b_1} \times \vec{b_2} = 4\hat{i} + 3\hat{j} - 12\hat{k}$

$\text{S.D.} = \left|\frac{(-2\lambda, -3, 12) \cdot (4, 3, -12)}{\sqrt{16 + 9 + 144}}\right|$

$\frac{-8\lambda - 9 - 144}{\sqrt{169}} = \pm 13$

$\Rightarrow 8\lambda + 153 = \pm 169$

$8\lambda = -153 \pm 169$

$8\lambda = -322$ or 16

So, $8\sum|\lambda| = \left|\frac{-322}{8} + \frac{16}{8}\right| 8$

$\boxed{8\sum|\lambda| = 306}$ Ans